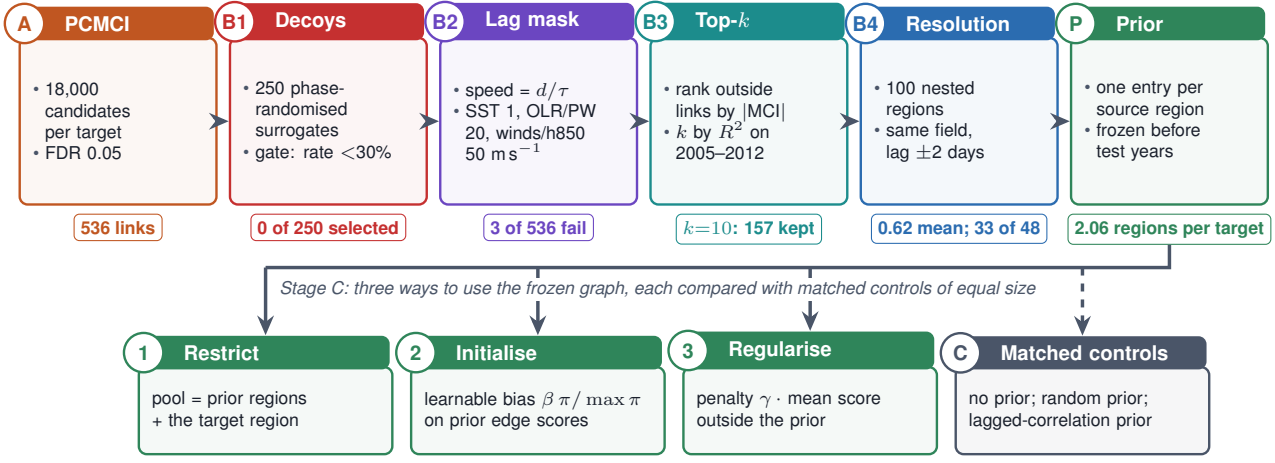

A
PCMCI
• 18,000 candidates per target
• FDR 0.05
536 links
B1
Decoys
• 250 phase-randomised surrogates
• gate: rate <30%
0 of 250 selected
B2
Lag mask
• speed = d/τ
• SST 1, OLR/PW 20, winds/h850 50 m s^{-1}
3 of 536 fail
B3
Top-k
• rank outside links by |MCI|
• k by R^2 on 2005–2012
k=10: 157 kept
B4
Resolution
• 100 nested regions
• same field, lag ±2 days
0.62 mean; 33 of 48
P
Prior
• one entry per source region
• frozen before test years
2.06 regions per target
Stage C: three ways to use the frozen graph, each compared with matched controls of equal size
1
Restrict
pool = prior regions + the target region
2
Initialise
learnable bias $\beta\,\pi/\max\pi$ on prior edge scores
3
Regularise
penalty $\gamma \cdot$ mean score outside the prior
C
Matched controls
no prior; random prior; lagged-correlation prior